

1. Give an example of each of the following, or explain why it is not possible.

a) A matrix that is similar to an identity matrix but not row equivalent to it.

Not possible. The only matrix similar to an identity matrix is itself, and it is also row equivalent to itself.

b) A matrix that is row equivalent to an identity matrix but not similar to it.

Any invertible matrix is row equivalent to an identity matrix, and none except the identity matrix is similar to it. So for example:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

c) a 3×7 matrix with nullity 4

One example is:

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

d) a 2×5 matrix with rank 3

Not possible. A matrix with 2 rows has rank at most 2.

e) A basis for $\mathbb{C}^3$ that contains no standard basis vectors

For example, the columns of $\begin{pmatrix} i & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1+i \end{pmatrix}$

f) A span in $\mathbb{C}^3$ that contains no standard basis vectors

Take the span of any vector that's not in the span of one of the standard basis vectors. For example,

$$\text{span} \left(\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\} \right)$$

g) A vector space with a pair of subspaces that are disjoint

Not possible. The intersection of any two subspaces of a vector space is nonempty, since it must contain the zero vector.

h) Four different examples of elementary matrices

$$\begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}, \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -7 & 1 \end{pmatrix}$$

i) a pair of matrices that are matrix equivalent but not row equivalent

$$\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \text{ and } \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

j) a pair of matrices that are row equivalent but not matrix equivalent

Not possible. For matrix equivalence, both row and column operations are allowed. So if two matrices are row equivalent, they are also matrix equivalent.

k) A non-identity matrix A such that $A^2 = A^{-1}$.

Note that the condition is equivalent to $A^3 = I$. Here are several examples:

$$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} \cos \frac{2\pi}{3} & -\sin \frac{2\pi}{3} & 0 \\ \sin \frac{2\pi}{3} & \cos \frac{2\pi}{3} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} \omega & 0 & 0 \\ 0 & \omega & 0 \\ 0 & 0 & \omega \end{pmatrix}, \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$$

The first is a permutation matrix where the three standard basis vectors move in a cycle of length 3. This holds over any field. The second is a real matrix that represents a counter-clockwise rotation by $\frac{2\pi}{3}$ about the z -axis in $\mathbb{R}^3$. The third is a diagonal matrix with diagonal entries equal to $\omega = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$, a third root of unity in $\mathbb{C}$; since $\omega^3 = 1$, the third power of the matrix is the identity. The last example is over $\mathbb{Z}/7\mathbb{Z}$, where $2^3 = 4^3 = 1$.

l) A linear transformation with domain $\mathbb{R}^3$ and codomain $\mathbb{C}^2$

Not possible. The scalars in two vector spaces must be the same in order to define a linear transformation between them.

m) a pair of similar matrices with different determinants

Not possible. Similar matrices have equal determinants, since $\det(QAQ^{-1}) = \det(Q)\det(A)\det(Q^{-1}) = \det(Q)\det(Q^{-1})\det(A) = \det(A)$.

n) a pair of matrices of different dimensions but the same determinant

Take for example the 2×2 and 3×3 identity matrices over $\mathbb{R}$. These both have the real number 1 as their determinant.

o) three distinct 2×2 matrices with integer entries whose inverses also have integer entries

It is sufficient (but not necessary) for the determinant to equal 1 (or -1), so:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 3 & 5 \\ 7 & 12 \end{pmatrix}, \begin{pmatrix} 10 & 9 \\ 11 & 10 \end{pmatrix}$$

Note that it is necessary and sufficient for the determinant to evenly divide all of the entries of the matrix.

p) A matrix with nullspace equal to its column space

This is only possible if we think of the domain and codomain as being identified; otherwise, the nullspace is a subspace of the domain and the column space is a subspace of the codomain, and so they cannot be equal. If we identify the domain and codomain (which forces the matrix to be square), an example is

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

since its column space is $\text{span} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and so is its nullspace.

q) A matrix with nullity equal to its rank

There are many possibilities, for instance:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

and any matrix row equivalent to these.

r) five different fields

$$\mathbb{R}, \mathbb{C}, \mathbb{Q}, \mathbb{Z}/5\mathbb{Z}, \mathbb{Z}/7\mathbb{Z}$$

s) a pair of 2×2 matrices that have the same determinant but are not similar

The matrices $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ both have determinant 1, but they are not similar since the only matrix similar to the identity is itself: $QIQ^{-1} = A$ implies $A = I$.

t) a vector space with exactly two subspaces

$\mathbb{R}^1$ is an example. Its only subspaces are the zero subspace and the whole vector space.

u) a vector space with exactly three subspaces

This is not possible. Any such vector space would necessarily be finite dimensional. It cannot have dimension 0 or 1, because those necessarily have 1 and 2 subspaces, respectively. But if a vector space has 2 or more dimensions, it has at least 2 1-dimensional subspaces, plus a 0-dimensional subspace and the whole space, which is more than three subspaces.

v) a vector space with exactly five subspaces

$(\mathbb{Z}/2\mathbb{Z})^2$ has exactly five subspaces: the zero subspace, the whole vector space, and the three 1-dimensional subspaces spanned by the three non-zero vectors: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

w) a matrix that is upper triangular but not diagonal

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

x) a matrix that is both symmetric and antisymmetric

A matrix A is symmetric if $A = A^T$ and it is antisymmetric if $A = -A^T$. (In each case, it follows that A is necessarily square.) Over infinite fields, the only examples are matrices where every entry is 0. Over $\mathbb{Z}/2\mathbb{Z}$, there are also examples like $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, since $-1 = 1$.

y) a linear transformation that is neither 1-to-1 nor onto

There are many examples. For instance, $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, as there is not a pivot in every column, and there is not a pivot in every row.

z) a map between two vector spaces that is not a linear transformation

Take the function $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ between vector spaces that sends every vector to the vector $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$. This is not a linear transformation, since it doesn't send the zero vector to the zero vector.

2. Explain what is wrong about the following definitions and correct them.

A **vector space** is a set V of vectors with the structure of a group under addition, a field $\mathbb{F}$ of scalars, and a multiplication of vectors by scalars that distributes over addition of vectors.

The underlying group of a vector space cannot be an arbitrary group; it must be *commutative*.

A **linear combination** of vectors is a sum of scalar multiples of the vectors.

The sum must have *finite* length.

The **span** of a set of vectors in a vector space is the set of all of their scalar multiples.

Not just their scalar multiples, but all of their *linear combinations*.

A **linear transformation** is a function between vector spaces over the same field that respects the operations of vector addition and vector multiplication: $L(v + w) = L(v) + L(w)$ and $L(vw) = L(v)L(w)$.

Not vector multiplication, but *scalar multiplication*: $L(\alpha v) = \alpha L(v)$

A **basis** of a vector space V is a set of vectors in V such that any vector v in V admits a representation as a linear combination of the vectors in the basis.

Not just a representation as a linear combination, but a *unique* such representation.

A matrix is in **row echelon form** if every non-zero row has its leading non-zero entry strictly to the right of the first non-zero entry in the previous row.

This, plus *all zero rows (if any) are below all non-zero rows*.

The **range** of a linear transformation $L : V \rightarrow W$ is the subset of W sent to the zero vector in W by L .

This is not the range, but rather the *nullspace* or *kernel*.

Two matrices A and B are **similar** (or **conjugate**) if there exist invertible matrices P and Q such that $A = PBQ$.

This is not the definition of similarity, but rather of *matrix equivalence*. For the definition of similarity, we must have $P = Q^{-1}$.

3. Complete the following:

a) State the Rank-Nullity Theorem. Illustrate the theorem with an example.

Rank-Nullity Theorem : For an $m \times n$ matrix A , $\text{rank}(A) + \dim(\text{Nul}(A)) = n$.

For instance, the matrix $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has two columns and its rank and nullity are each 1.

b) Give four examples of conditions that are equivalent to a square matrix being invertible.

Refer to the Invertible Matrix Theorem. For example:

A is row equivalent to an identity matrix.

$Ax = b$ has a unique solution for all b in $\mathbb{F}^n$.

The linear transformation T represented by A is onto.

$\det(A) \neq 0$

c) State Cramer's rule. Illustrate the rule with an example.

Cramer's rule: For an invertible matrix A , the k th entry x_k of the solution of the equation $Ax = b$ is given by the formula

$$x_k = \frac{\det B_k}{\det A},$$

where the matrix B_k is obtained from A by replacing column number k of A by the vector b .

So for instance, the matrix equation $\begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix} x = \begin{pmatrix} 7 \\ 8 \end{pmatrix}$ is given by

$$\begin{pmatrix} \left| \begin{array}{cc|c} 7 & 4 & 3 \\ 8 & 2 & 1 \end{array} \right| \\ \left| \begin{array}{cc|c} 3 & 4 & 7 \\ 1 & 2 & 8 \end{array} \right| \\ \left| \begin{array}{cc|c} 3 & 7 & 1 \\ 1 & 8 & 8 \end{array} \right| \\ \left| \begin{array}{cc|c} 3 & 4 & 1 \end{array} \right| \end{pmatrix} = \begin{pmatrix} \frac{-18}{2} \\ \frac{17}{2} \end{pmatrix} = \begin{pmatrix} -9 \\ \frac{17}{2} \end{pmatrix}$$

And indeed we have $\begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} -9 \\ \frac{17}{2} \end{pmatrix} = \begin{pmatrix} 7 \\ 8 \end{pmatrix}$

4. Prove each of the following:

a) Using the vector space axioms, show that $0\mathbf{v} = \mathbf{0}$ for any vector $\mathbf{v}$

$$0\mathbf{v} = (0 + 0)\mathbf{v} = 0\mathbf{v} + 0\mathbf{v}$$

This uses the property that in a field, 0 plus anything is that thing; and also the distributivity of scaling over vector addition.

Now add the additive inverse of $0\mathbf{v}$ to both sides, and simplify:

$$0\mathbf{v} = 0\mathbf{v} + 0\mathbf{v} \implies 0\mathbf{v} + -0\mathbf{v} = 0\mathbf{v} + 0\mathbf{v} + -0\mathbf{v} \implies \mathbf{0} = 0\mathbf{v} + \mathbf{0} \implies \mathbf{0} = 0\mathbf{v}$$

Here we have used the associativity of vector addition, the defining property of additive inverses, and the property that any vector plus the zero vector is that vector.

b) The intersection of two subspaces is a subspace.

Let U and W be two subspaces of a vector space V . Since each of U and W contain the zero vector, so does $U \cap W$. Additionally, any linear combination of vectors in $U \cap W$ lies in U , since all of the vectors in $U \cap W$ lie in U , and U as a subspace is closed under linear combinations. And similarly for W . Therefore any linear combination of vectors in $U \cap W$ lies in both U and W and thus in $U \cap W$, that is, $U \cap W$ is closed under linear combinations. Since these two properties are satisfied, $U \cap W$ is a subspace.

c) Using the defining properties of the determinant, show that a matrix with linearly dependent columns has determinant 0.

If a matrix A has linearly dependent columns, column reducing it results in a matrix with a zero column. This matrix is column equivalent to a matrix with two equal columns. But then by the alternating property of the determinant, this matrix must have determinant 0. Therefore the original matrix A also has 0 for its determinant, since column operations are reversible and either preserve the determinant or scale the determinant by a non-zero constant.

d) If a matrix A represents a one-to-one linear transformation, then A^T represents an onto linear transformation.

Since A represents a one-to-one linear transformation, it has a pivot in every column. But then A^T has a pivot in every row, and so represents an onto linear transformation.

5. Always, sometimes, or never? If it is always or never true, give a proof. If it is sometimes true, give an example where it's true and where it's false.

a) An invertible matrix has determinant 1

Sometimes. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is invertible and has determinant 1, but $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ is also invertible but has determinant 4.

b) A lower triangular matrix is in RREF

Sometimes. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is LT and RREF, while $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ is LT but not RREF.

c) An upper triangular matrix is invertible

Sometimes. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is UT and invertible, while $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ is UT but not invertible.

d) Three vectors in $\mathbb{R}^3$ are linearly independent

Sometimes. The columns of $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ are linearly independent, but those of $\begin{pmatrix} 1 & 2 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix}$ are not.

e) The vectors in a basis for a vector space V span a subspace of V .

Always, since V itself is a subspace of V , and any basis must span its vector space.

f) The sum of two matrices has fewer entries than either of the two matrices.

Never. Whenever matrix addition is defined, the two matrices are both $m \times n$ and so is the sum.

g) The product of two matrices has fewer entries than either of the two matrices.

Sometimes. For $\begin{pmatrix} 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \\ 9 \end{pmatrix} = (50)$, the product has one entry and the two matrices each have three entries. On the other hand, the product of two $n \times n$ matrices is also $n \times n$ and so does not have fewer entries.

h) An invertible matrix has an LU decomposition.

Sometimes. An invertible matrix that can be put into a row echelon form matrix (note: REF is not unique) without swapping rows or adding a multiple of a lower row to a higher one has an LU decomposition. For example:

$$\begin{pmatrix} 3 & 9 \\ 4 & 5 \end{pmatrix} \xrightarrow{\frac{1}{3}R_1} \begin{pmatrix} 1 & 3 \\ 4 & 5 \end{pmatrix} \xrightarrow{R_2-4R_1} \begin{pmatrix} 1 & 3 \\ 0 & -7 \end{pmatrix}$$

By rewriting the reduction process as applying elementary matrices, we have:

$$\begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 9 \\ 4 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 0 & -7 \end{pmatrix}$$

or

$$\begin{pmatrix} 3 & 9 \\ 4 & 5 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 0 & -7 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 0 & -7 \end{pmatrix}$$

This yields the desired LU decomposition.

However, if a row swap or adding a multiple of a lower row to a higher one is required to bring the matrix into row echelon form, we cannot derive an LU decomposition. Such is the case for the following matrix: $\begin{pmatrix} 0 & 4 \\ 1 & 6 \end{pmatrix}$

When this occurs, we can always permute the rows first and then proceed to get an LU decomposition on the result. Thus every invertible matrix has a PLU decomposition.

i) Two distinct subspaces have different dimensions.

Sometimes. For instance, in $\mathbb{R}^2$, the whole vector space and the zero subspace are distinct and have different dimensions. On the other hand, the spans of the two standard basis vectors are distinct but each have dimension 1.

j) When v and w are linearly independent, so are $v + w$ and $v - w$.

Always. Consider $\text{span}(\{v + w, v - w\})$. It contains $\frac{1}{2}(v + w) + \frac{1}{2}(v - w) = v$, and similarly it contains w . Thus the dimension of this span is 2, since it is at most 2 and contains the span

of v and w , which has dimension 2. Therefore $v + w$ and $v - w$ are linearly independent.

k) A matrix is left invertible but not right invertible.

Sometimes. A matrix whose linear transformation is 1-to-1 but not onto is left invertible but not right invertible. However, there are left invertible matrices that are also right invertible, namely all invertible matrices.

l) A matrix is row equivalent to a matrix that is in reduced row echelon form.

Always. This is one of our important theorems. In fact, a matrix is row equivalent to exactly one matrix that is in RREF.

6. For these two pairs of matrices, decide whether or not they are row equivalent. Explain your thinking.

$$A_1 = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & -1 \end{pmatrix} \quad \& \quad A_2 = \begin{pmatrix} 1 & 0 & 0 \\ 4 & 9 & 0 \\ 6 & -3 & 10 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 5 & 6 \\ 0 & 0 & 9 \end{pmatrix} \quad \& \quad B_2 = \begin{pmatrix} 1 & 7 & 2 \\ 1 & 10 & 2 \\ 1 & -5 & 2 \end{pmatrix}$$

Both A_1 and A_2 have 3 pivot columns—they are each triangular with non-zero entries on the diagonal. Therefore they are both row equivalent to the identity matrix, and they're therefore row equivalent to each other.

Similarly, B_1 is also rank 3. But B_2 is rank 2: its first and third columns are not linearly independent. Since row operations preserve rank, B_1 and B_2 are not row equivalent.

7. Give an example of a matrix that is similar to $\begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix}$ but not equal to it.

It is enough to choose some invertible matrix Q , to compute $Q \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix} Q^{-1}$, and then check that the resulting matrix is distinct from the given one.

For instance

$$\begin{pmatrix} 1 & 5 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & -5 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -5 \\ 0 & 3 \end{pmatrix}$$

8. Give a basis for each of the four fundamental subspaces of the following matrix: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

Let A be the given matrix. The four fundamental subspaces are the column space and nullspace of A , and the same subspaces for A^T . Note that for this particular A , $A = A^T$.

A basis for the column space of A is $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$.

A basis for the nullspace is $\left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$.

A basis for the column space of A^T is also $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$.

And a basis for the nullspace of A^T is also $\left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$.

9. Give two different bases for the vector space $\mathcal{P}_3$ of real polynomials of degree at most 2. For each basis, give the matrix for the linear transformation that computes the derivative map $\mathcal{P}_3 \rightarrow \mathcal{P}_3$. Using each matrix, find a basis for the nullspace of the map.

One basis for $\mathcal{P}_3$ is $B_1 = \{1, x, x^2\}$, so that 1 is represented by $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, x is represented by

$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, and x^2 is represented by $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

Then the derivative map $\mathcal{P}_3 \rightarrow \mathcal{P}_3$ is given in the basis B_1 by

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

A basis for the nullspace is $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}$; note that all (and only) constant polynomials are in kernel.

Another basis for $\mathcal{P}_3$ is $B_2 = \{x + 1, x - 1, x^2 + x + 1\}$.

The derivative map $\mathcal{P}_3 \rightarrow \mathcal{P}_3$ is given in the basis B_2 by

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

After row reducing we find that now a basis for the nullspace is $\left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \right\}$; note that again all (and only) constant polynomials are kernel.

10. For any n , define the $n \times n$ checkerboard matrix C_n as follows: $c_{i,j} = 1$ if $i + j$ is even, and 0 otherwise. For example, here is C_5 :

$$\begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

Find a formula for $\det C_n$.

When n is 1, $C_1 = (1)$, and we have $\det C_1 = 1$.

When n is 2, $C_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, and we compute that $\det C_2 = 1$.

When $n > 2$, we observe that the columns of C_n are not linearly independent, as the first and third columns are equal. We have as a consequence that $\det C_n = 0$ for all $n > 2$.